

Example: consider a 3-stage rocket

$$\epsilon_1 = 0.0765, \epsilon_2 = 0.114, \epsilon_3 = 0.111 \quad (\text{structural coeff})$$

$$I_{sp1} = 304 \text{ s}, I_{sp2} = 421 \text{ s}, I_{sp3} = 421 \text{ s} \quad (I_{sp})$$

$$\Delta u_{b,imp,tot} = 12.4 \frac{\text{km}}{\text{s}} \quad (\Delta u_{b,imp,tot})$$

optimize the rocket for Maximum $\left(\frac{M_L}{M_{o1}}\right)$

Solution: Use MATLAB for solving (E), then get λ_i from (D)

$$\alpha_g = 0.0038966$$

$$\lambda_1 = 0.8142, \lambda_2 = 0.2514, \lambda_3 = 0.2421$$

$$\left(\frac{M_L}{M_{o1,opt}}\right) = 0.01758 \rightarrow \Delta u_{b,imp,1} = 2121.5 \frac{\text{m}}{\text{s}}$$

$$\Delta u_{b,imp,2} = 5084.2 \frac{\text{m}}{\text{s}}$$

(P-10)

$$\Delta u_{b,imp,3} = 5194 \frac{\text{m}}{\text{s}} \rightarrow \Delta u_{b,imp,tot} = 12.4 \text{ km/s}$$

"This is optimization of Saturn V rocket"

$$M_0 = 2,902,000 \text{ kg}$$

$$M_L (\text{mass of Apollo 11}) = 47,000 \text{ kg}$$

$$\frac{M_L}{M_0} = 0.0162 \rightarrow \text{our optimization gave } \frac{M_L}{M_0}_{\text{opt}} = 0.01758$$

"Similarly Staged Rocket"

A rocket with stages having equal ϵ_i & I_{sp_i} is called similarly staged. From Eqⁿ (D), this rocket has equal λ_i as well. For this rocket, Eqⁿ (E) simplifies to:

$$n g_e I_{sp} \ln \left(\frac{1+\lambda}{\epsilon+1} \right) = \Delta u_{b, \text{simp}} \quad (F)$$

number of stages

$$\lambda = \lambda_1 = \lambda_2 = \lambda_3 = \dots$$

$$I_{sp} = I_{sp_1} = I_{sp_2} = I_{sp_3} = \dots$$

$$\epsilon = \epsilon_1 = \epsilon_2 = \epsilon_3 = \dots$$

λ can be found from Eqⁿ (F), without having to find α , if needed, α can be found from (D) after λ has been determined:

$$\lambda = \frac{\epsilon}{\alpha g_e I_{sp} (1-\epsilon) - 1}$$

Example of Three-stage rocket previously discussed

$$M_{o_1} = 2.5 \times 10^6 \text{ kg}, \Delta u_b = 11023 \frac{\text{m}}{\text{s}}, I_{sp} \text{ (for all stages)} = 370 \text{ (s)},$$

$$M_s = 145 \times 10^3 \text{ kg}, M_p = 2.287 \times 10^6 \text{ kg}$$

$$\epsilon = \epsilon_1 = \epsilon_2 = \epsilon_3$$

Find λ of $\frac{M_L}{M_{o_1}}$:

Total structural mass

Total propellant mass

$$\epsilon_1 = \frac{M_{s_1}}{M_{s_1} + M_{p_1}} = \epsilon_2 = \frac{M_{s_2}}{M_{s_2} + M_{p_2}} = \epsilon_3 = \frac{M_{s_3}}{M_{s_3} + M_{p_3}}$$

$$= \epsilon = \frac{M_{s_1} + M_{s_2} + M_{s_3}}{(M_{s_1} + M_{s_2} + M_{s_3}) + (M_{p_1} + M_{p_2} + M_{p_3})} = \frac{M_{s_{tot}}}{M_{s_{tot}} + M_{p_{tot}}} \Rightarrow$$

$$\epsilon = \frac{145 \times 10^3}{145 \times 10^3 + 2.287 \times 10^6} = 0.05962$$

compute λ from (F)

$$n g_e I_{sp} \ln\left(\frac{1+\lambda}{\epsilon+\lambda}\right) = \Delta u_{b,imp} \quad (F)$$

$$3 \times 9.81 \times 370 \times \ln\left(\frac{1+\lambda}{0.05962+\lambda}\right) = 11023$$

$$\Rightarrow \lambda = 0.4772$$

$$\frac{M_L}{M_{o_1}} = \prod_{i=1}^n \frac{\lambda_i}{1+\lambda_i} \longrightarrow \text{for similarly staged rocket}$$

this simplifies to:

$$\frac{M_L}{M_{o_1}} = \left(\frac{\lambda}{1+\lambda}\right)^n$$

(P-2)

"Rockets & Thermodynamics"

In this Example: $\frac{M_L}{M_{o_1}} = \left(\frac{\lambda}{1+\lambda} \right)^3 = 0.0337 > 0.0294$

Note: Similarly staged rockets has maximum $\frac{M_L}{M_{o_1}}$ even more than optimized rockets.
